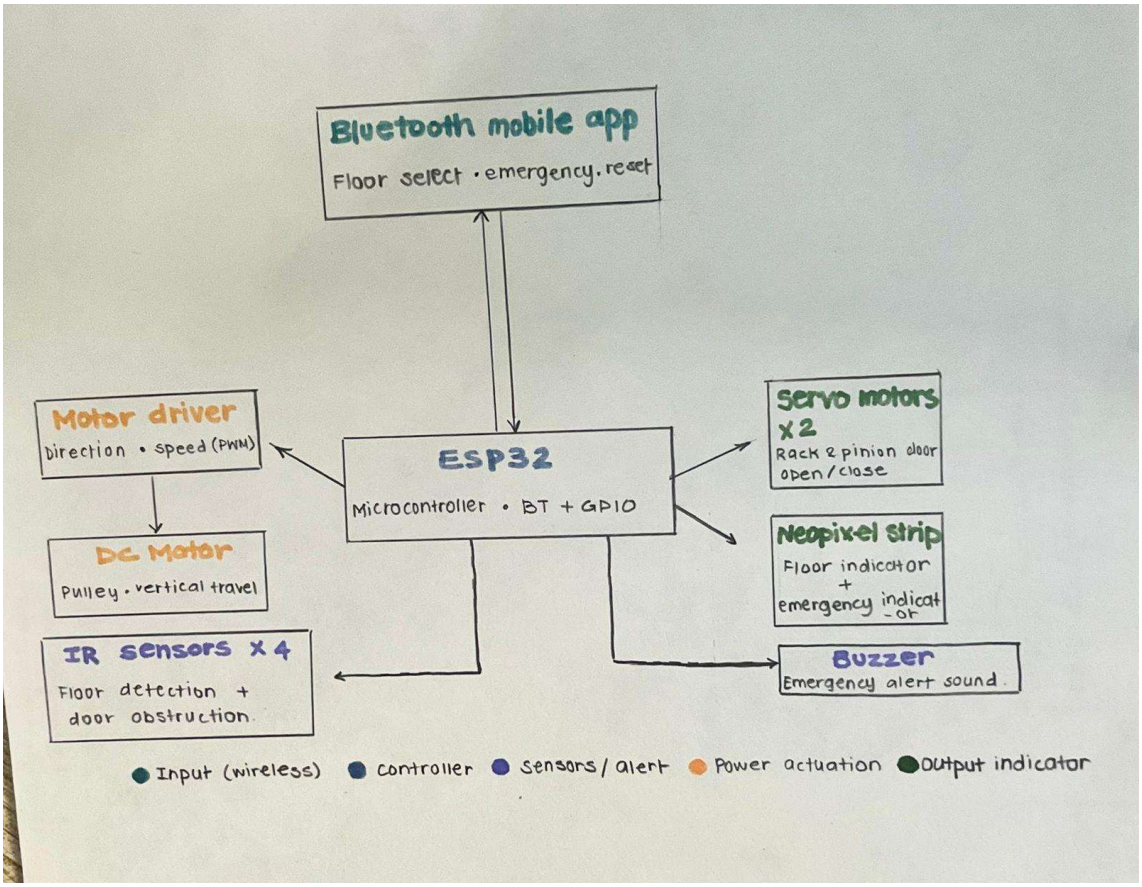
Electronics Documentation

1. **Electronics Block Diagram**



2. **Components Table**

Component	**Quantity**	**Purpose**	**Justification**
Servo Motor	2	To drive the rack and pinion mechanism for opening and closing the elevator doors	Provides controlled angular motion which can be converted into precise linear movement for smooth and synchronized door operation
IR Sensors	4	To detect elevator position at each floor and detect obstruction between doors	Allows accurate floor detection and ensures safety by preventing door closure when obstructed
ESP32 Microcontroller	1	To control all system operations and handle communication between components	Offers integrated Bluetooth capability and sufficient GPIO pins for managing multiple inputs and outputs

Component	Quantity	Purpose	Justification
DC Motor	1	To drive the pulley system for vertical movement of the elevator	Provides continuous rotational motion required for lifting and lowering the elevator car
Motor Driver	1	To control the direction and speed of the DC motor	Required to interface the DC motor with the microcontroller and handle higher current safely
Bluetooth Mobile Application	1	To send control commands (floor selection, emergency, reset) to the system	Enables wireless user interaction and enhances usability of the system